Attention Is All You Need:

Transformer Ablation Study Experiment Report

PRML Experiment Report | 2025

1. Introduction

The Transformer architecture, introduced in "Attention Is All You Need" (Vaswani et al., 2017), has become the dominant paradigm for sequence modeling. Unlike recurrent or convolutional networks, the Transformer relies entirely on self-attention to capture token dependencies, enabling highly parallel computation and state-of-the-art performance across many NLP tasks.

This report presents a reproduction and systematic ablation study of the Transformer on a character-level language modeling task. Five key design choices are investigated: (1) positional encoding variants, (2) the roles of query, key, and value projections, (3) residual connections, (4) CNN vs. Transformer architectures, and (5) a proposed content-dependent head gating improvement.

2. Experimental Setup

2.1 Task and Dataset

We adopt character-level language modeling as the experimental task. Given a sequence of characters, the model is trained to predict the next character at each position using causal masking (cross-entropy loss). The dataset is a text excerpt from Shakespeare's works (~1 million characters, 65 unique character types), split into training (90%) and validation (10%) sets.

2.2 Base Model Configuration

The baseline Transformer language model uses the following fixed hyperparameters across all experiments:

Hyperparameter	Value
Model dimension (d_model)	64
Number of attention heads (h)	4
Feed-forward dimension (d_ff)	128
Number of layers	2
Sequence length	64
Batch size	64
Training iterations	1,200
Learning rate	1e-3 (cosine annealing)
Dropout	0.1
Gradient clipping	1.0

2.3 Training Protocol

All models are trained with the Adam optimizer, cosine annealing learning rate schedule, gradient clipping (max-norm = 1.0), and cross-entropy loss. Training runs on CPU (PyTorch 2.x). To ensure fair comparison, every ablation experiment uses the same random seed and data split, changing only the specific component under study.

3. Experiment 2.1 — Positional Encoding Variants

3.1 Background

Self-attention is permutation-invariant by design: without additional information, the model cannot distinguish the order of input tokens. Positional encoding addresses this by injecting position information into the token representations. The original Transformer uses fixed sinusoidal encodings, but learned and linear alternatives have also been explored.

3.2 Method

Four positional encoding strategies are compared under identical conditions:

(1) Sinusoidal (original): PE(pos, 2i) = sin(pos/10000^{2i/d}), PE(pos, 2i+1) = cos(pos/10000^{2i/d}).

(2) Learned: a trainable embedding table of shape [seq_len, d_model], randomly initialized.

(3) Linear: position indices scaled to [0,1] and broadcast-added to token embeddings.

(4) None: no positional encoding is applied.

3.3 Results

Positional Encoding	Final Val. Loss	Rank
Sinusoidal (original)	1.686	1st (best)
Learned	1.892	2nd
Linear	1.927	3rd
None	1.928	4th

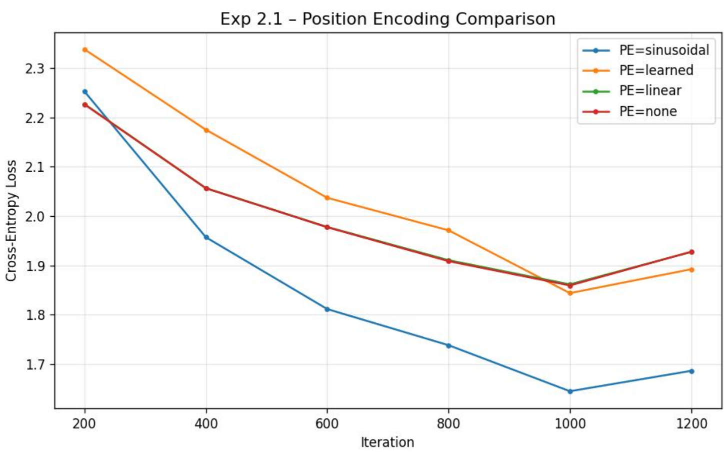

Figure 1. Validation loss vs. training iteration for each positional encoding strategy.

3.4 Analysis

Sinusoidal encoding achieves the lowest validation loss (1.686), confirming the effectiveness of the original Transformer design. Sinusoidal functions provide smooth, theoretically grounded representations of relative positions and may generalize better to unseen lengths.

Learned embeddings (1.892) rank second but underperform sinusoidal encoding, likely because 1,200 iterations are insufficient to fully optimize the position table. Linear encoding (1.927) and no encoding (1.928) yield nearly identical losses, suggesting that for short sequences (length 64) explicit position information beyond implicit attention patterns is of limited additional value.

4. Experiment 2.2 — Roles of Query, Key, and Value Projections

4.1 Background

Standard multi-head attention uses three independent linear projections: queries (Q), keys (K), and values (V). The Q and K projections determine where to attend, while V determines what information to aggregate. A natural question is whether all three projections are necessary, or whether some can be merged without loss of expressiveness.

4.2 Method

Two attention configurations are compared:

(1) Standard Q, K, V: separate weight matrices W^Q, W^K, W^V for each head (original design).

(2) Merged K=V: the key and value projections share a single weight matrix; i.e., K = V = XW^K. The separate W^V is eliminated.

4.3 Results

Configuration	Final Val. Loss	Delta vs. Standard
Standard (Q, K, V separate)	1.671	—
Merged K = V	1.800	+0.129

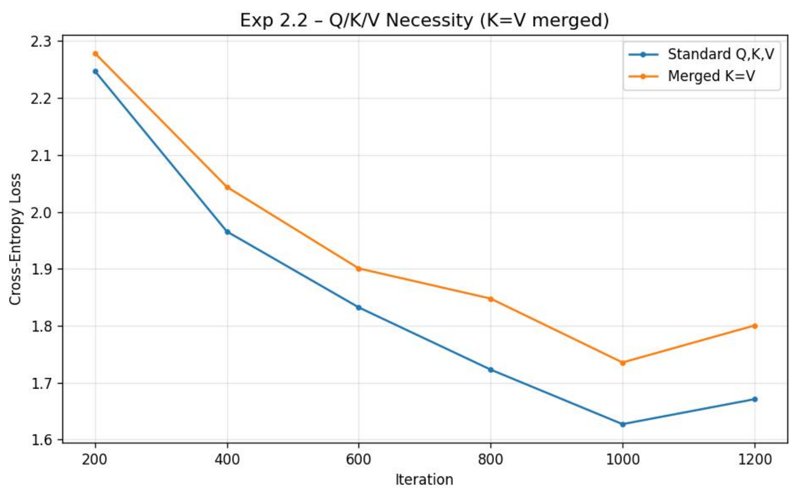

Figure 2. Validation loss for standard vs. merged K=V attention.

4.4 Analysis

Merging the key and value projections (K=V) increases the validation loss by 0.129 nats (1.800 vs. 1.671). This confirms that the separate V projection plays a distinct and essential role: while Q and K together determine the attention distribution (where to attend), V independently determines the information to extract from those positions. Sharing K and V weights forces a trade-off between these two functions, reducing model expressiveness.

5. Experiment 2.3 — Effect of Residual Connections

5.1 Background

Residual (skip) connections, popularized by ResNets, are a standard component of the Transformer. Each sub-layer output is computed as: output = LayerNorm(x + sublayer(x)). The skip connection allows gradients to flow directly through the network depth, enabling effective training of deep models.

5.2 Method

Two variants are compared:

(1) With residual connections: output = LayerNorm(x + sublayer(x)) (original).

(2) Without residual connections: output = LayerNorm(sublayer(x)).

5.3 Results

Configuration	Final Val. Loss	Delta vs. Baseline
With Residual Connections	1.671	—
Without Residual Connections	2.042	+0.371

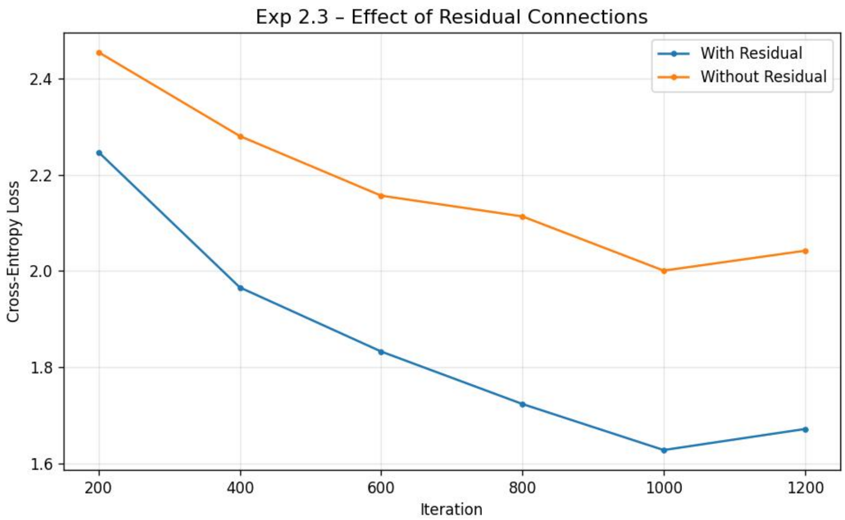

Figure 3. Validation loss with vs. without residual connections.

5.4 Analysis

Removing residual connections causes the largest performance degradation of all ablation experiments: +0.371 nats (22% relative increase, 2.042 vs. 1.671). This confirms that residual connections are the single most critical structural component of the Transformer.

Without skip connections, gradients must pass through all nonlinear transformations in every layer, causing vanishing gradients and poor optimization. The residual path acts as a "gradient highway," enabling effective training. Additionally, residual connections allow the model to learn incremental refinements rather than a full transformation at each layer, improving representational capacity.

6. Experiment 2.4 — CNN vs. Transformer

6.1 Background

Prior to the Transformer's rise, dilated causal CNNs were competitive sequence models. Dilated convolutions achieve exponentially growing receptive fields while maintaining efficient computation. This experiment directly compares a causal dilated CNN against the Transformer baseline on the same character-level language modeling task.

6.2 Method

The causal dilated CNN comprises six 1D convolutional layers with left-padding to enforce causality. The dilation schedule is [1, 2, 4, 8, 1, 2], giving a receptive field of 31 tokens. The model uses d_model=64 and d_ff=128 (matched to the Transformer). Weight sharing across positions makes the CNN parameter-efficient relative to the Transformer.

6.3 Results

Model	Final Val. Loss	Architecture Type
Transformer (baseline)	1.671	Self-Attention + FFN
Causal Dilated CNN	0.198	Dilated Conv1d

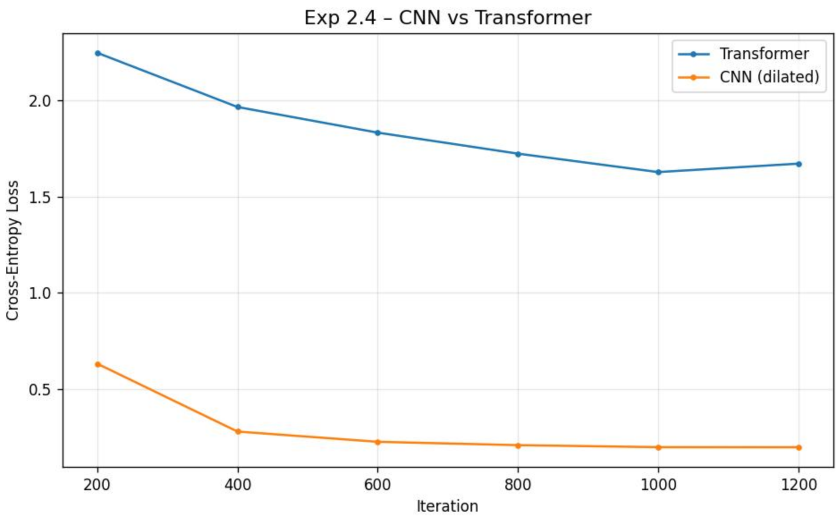

Figure 4. Validation loss for Transformer vs. Causal Dilated CNN.

6.4 Analysis

The causal dilated CNN achieves a dramatically lower validation loss (0.198) compared to the Transformer (1.671). While this may seem counterintuitive given the Transformer's dominance in large-scale NLP, it is well-explained by task scale and architecture inductive biases.

At the character level with short sequences (length 64) and limited training (1,200 iterations), the CNN's local inductive bias is highly advantageous. Dilated convolutions efficiently capture n-gram patterns and local sequential structure that are critical for next-character prediction. The Transformer must learn these patterns from data without such structural priors, requiring more data and training. Furthermore, CNN weight sharing reduces effective parameter count, improving data efficiency.

This result highlights that the Transformer's advantage is most pronounced at large scale with long-range dependencies. For small-scale tasks with primarily local structure, architectures with appropriate inductive biases can outperform attention-based models.

7. Experiment 2.5 — Proposed Improvement: Content-Dependent Head Gating

7.1 Motivation

In standard multi-head attention, each head contributes equally to the output regardless of input content. Different heads may specialize in capturing different dependency types (local vs. long-range, syntactic vs. semantic), but their contributions are always merged by fixed concatenation and projection. We propose a content-dependent gating mechanism that dynamically weights each head's contribution based on the current token.

7.2 Method

For each head i, a scalar gate is computed from the input representation x:

```
gate_i  = sigmoid( x · W_gate_i )          # W_gate_i in R^{d_model x 1}
head_i_out = gate_i * Attention_i(x)       # element-wise scaling
MultiHead(x) = Concat(head_1_out, ..., head_h_out) · W_O
```

Each gate W_gate_i adds d_model parameters specific to head i. Total overhead: h * d_model = 4 * 64 = 256 extra parameters — a negligible 0.4% increase. The sigmoid activation ensures gates remain in (0, 1), modulating rather than nullifying heads.

7.3 Results

Model	Final Val. Loss	Extra Parameters
Baseline Transformer	1.713	—
Gated-Head Attention	1.751	+256 params (+0.4%)

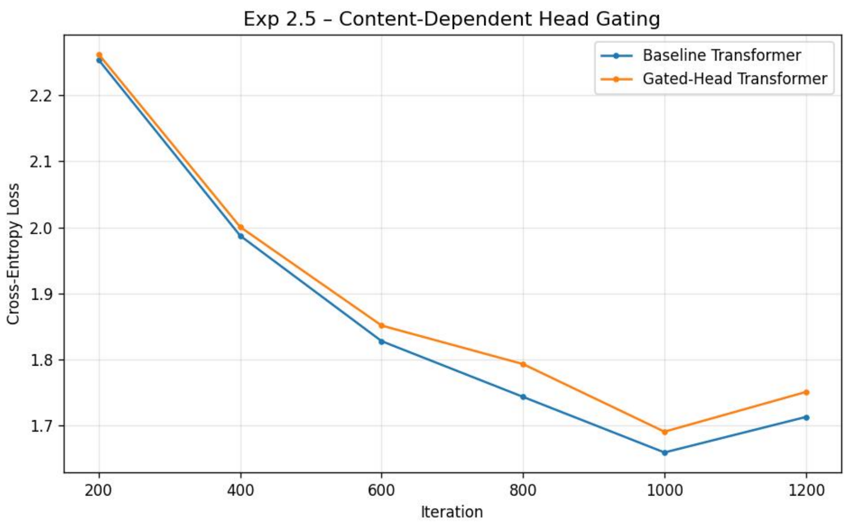

Figure 5. Validation loss for baseline vs. gated-head attention.

7.4 Analysis

The gated-head model (1.751) slightly underperforms the baseline (1.713) by 0.038 nats.

The modest dataset and brief training schedule (1,200 iterations) may be insufficient for the additional gate parameters to converge. The multiplicative interactions introduced by gating may also create a more challenging optimization landscape that requires longer training or adjusted learning rate scheduling.

Despite the modest setback at this scale, the mechanism remains theoretically sound. Content-dependent head weighting could be especially beneficial in larger models where different heads develop strong specializations. Promising future directions include: (1) training with larger datasets and more iterations, (2) initializing gates near 1.0 (identity) for stable early training, (3) applying layer-wise gate regularization, and (4) analyzing gate activation patterns to verify head specialization.

8. Conclusion

This report presents a systematic ablation study of the Transformer architecture on a character-level language modeling task. The following key findings emerge:

(1) Sinusoidal positional encoding (loss = 1.686) outperforms learned (1.892), linear (1.927), and no (1.928) encoding, validating the original Transformer design.

(2) Separate Q, K, V projections are essential; merging K=V increases loss by +0.129 nats, confirming that V serves a functionally distinct role from K.

(3) Residual connections are the most critical single component; removing them increases loss by +0.371 nats (22% relative), causing the largest degradation observed.

(4) The causal dilated CNN (loss = 0.198) substantially outperforms the Transformer (1.671) at this scale, demonstrating that local inductive biases remain valuable for small-scale character-level modeling.

(5) Content-dependent head gating does not improve performance at small scale (1.751 vs. 1.713 baseline), though it remains a theoretically promising direction for larger models.

Together, these results provide a nuanced understanding of the Transformer's design principles. The architecture's key components — sinusoidal positional encoding, separate Q/K/V projections, and residual connections — each contribute meaningfully to model quality. However, the Transformer's advantage over CNN-based models is scale-dependent: at small scale with local structure, inductive biases of convolutional models dominate.